

- Q.19 What is the Central Limit Theorem? How does it help in understanding simulation results? (CO3)
- Q.20 What is skewness in simulation results, and why should we care about it? (CO3)
- Q.21 Write a note on practical applications of PERT. (CO4)
- Q.22 List some good reasons to use simulation software in manufacturing? (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 How do the Runge-Kutta and Monte Carlo methods differ in simulation? Provide a manufacturing example for each. (CO1)
- Q.24 Describe in detail; how queuing models are used to simulate manufacturing systems. Include few real life example. (CO3)
- Q.25 What are the key features of GPSS? How is it used to simulate manufacturing system? Give a basic example. (CO4)

No. of Printed Pages : 4
Roll No.

222864C/212864C

6th Sem. / Automation & Robotics

Subject : Modeling, Simulation and Analysis of Manufacturing Systems

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is usually the first step in a simulation study?
a) Collecting output data (CO1)
b) Defining problem and goals
c) Running the model d) Validating results
- Q.2 What is a limitation of system simulation? (CO1)
a) Needs high computational power
b) Always faster than real tests
c) Can't model discrete systems
d) No data collection needed
- Q.3 What is a key feature of Monte Carlo methods in simulation? (CO1)
a) Use fixed calculations
b) Use random sampling
c) Only for continuous systems
d) Need physical prototypes

- Q.4 Which test checks random number independence in simulation? (CO2)
- Chi-square test
 - Normal distribution test
 - Standard error test
 - Autocorrelation test
- Q.5 What best describes pseudorandom numbers?(CO2)
- Generated by natural events
 - Created using algorithms
 - Perfectly uniform in distribution
 - Not suitable for simulation
- Q.6 What defines a stochastic activity in simulation?
- Fixed time and result (CO2)
 - Involves randomness
 - Only for continuous systems
 - Uses only analytical methods

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 PERT is used to calculate the critical path duration only. (True/False) (CO4)
- Q.8 Queue discipline is a key component in queuing simulations. (True/False) (Co3)

- Q.9 GPSS block diagrams visually represent entity flow in a system. (True/False) (CO4)
- Q.10 Skewness measures the _____ of a data distribution. (CO3)
- Q.11 In PERT, an event marks the _____ or _____ of an activity. (CO4)
- Q.12 What types of block in GPSS creates entities in a simulation? (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Compare and contrast continuous and discrete systems? Give one example of each. (CO1)
- Q.14 What is the Monte Carlo method? How can it be used in a simple manufacturing situation? (CO1)
- Q.15 What are the Binomial and Poisson distributions used for in simulation? Discuss. (CO2)
- Q.16 What are the main parts of a queuing system? Explain in brief. (CO3)
- Q.17 Differentiate between time-oriented and event-oriented simulations with suitable examples?(CO3)
- Q.18 Write a note on hash algorithm? (CO2)